

8.3 UNIT TAX & SUBSIDY

Multiple Choice Questions

1990/CE/II/51

Suppose the government increases the tax on cigarettes by \$2. Consequently, the price of each packet of cigarettes increases by \$1.5. Which of the following statements is true?

- A. The producers bear 1/4 of the tax
- B. The demand for cigarettes is elastic
- C. The supply of cigarettes is elastic
- D. The tax is progressive in nature

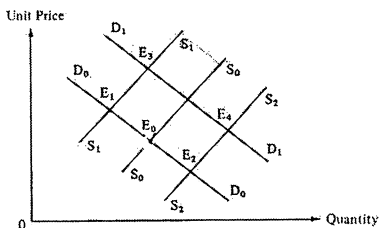
1991/CE/II/58

Suppose the government raises the first registration tax on newly-imported cars by \$ 10 000. Which of the following is FALSE?

- A. The market price of all newly-imported cars will increase by \$10 000.
- B. The market price of second-hand cars will increase.
- C. The tax burden can be shifted to the consumers.
- D. The number of cars newly imported will decrease.

1992/CE/II/25

The diagram below shows the demand and supply of Good X. The original equilibrium point is E_0 .



If the government imposes an excise tax on the good, the equilibrium point will change from E_0 to

- A. E_1
- B. E_2
- C. E_3
- D. E_4

1993/CE/II/29

A tax is imposed on a commodity. The tax burden on the consumers is relatively smaller if the commodity has an _____ demand and an _____ supply.

- A. elastic elastic
- B. elastic inelastic
- C. inelastic elastic
- D. inelastic inelastic

For Question No.8 and 9, consider the following supply and demand schedule of wine in a certain country:

Price (\$)	Quantity demanded (Units)	Quantity supplied (Units)
30	90	60
32	80	65
34	70	70
36	60	75
38	50	80
40	40	85

1994/CE/II/08

Suppose initially the wine market is in equilibrium. If the government imposes a tax of \$6 for each unit of wine sold, the new equilibrium price will be

- A. \$32
- B. \$36
- C. \$38
- D. \$40

1994/CE/II/09

Government revenue from the above tax will be _____.

- A. \$240
- B. \$300
- C. \$360
- D. \$420

1996/CE/II/07

Price (\$)	Quantity demanded (units)	Quantity supplied (units)
6	60	40
8	55	45
10	50	50
12	45	55
14	40	60

Refer to the above information about a good. What will be the consumer tax burden if a per-unit tax of \$ 4 is imposed?

- A. \$90
- B. \$100
- C. \$110
- D. \$180

1998/CE/II/41

Suppose the sales tax rate of a good is raised. Which of the following statement is correct?

- A. The total expenditure on the good will increase.
- B. There will be inflation.
- C. The total revenue net of tax to sellers will fall if the supply curve is upward sloping.
- D. Income will be redistributed from the poor to the rich.

Refer to the following supply and demand schedule for good Y and answer questions 8 and 9

Unit price (\$)	12	11	10	9	8	7
Quantity demanded (Units)	40	60	80	100	120	140
Quality supplied (units)	280	240	200	160	120	80

1999/CE/II/08

If the government imposed a \$3 per unit tax on Y, what would the new total sales revenue be?

- A. \$360
- B. \$800
- C. \$960
- D. \$1 320

1999/CE/II/09

The ratio of the buyers' tax burden to the sellers' tax burden is

- A. 1 : 1
- B. 2 : 1
- C. 1 : 2
- D. 3 : 0

2001/CE/II/08

Suppose a per unit sales tax is imposed on good X. The sellers would bear a relatively smaller proportion of the tax burden if the demand for the good is _____ and the supply is _____.

- A. elastic elastic
- B. elastic inelastic
- C. inelastic elastic
- D. inelastic inelastic

2003/CE/II/06

If a country reduces a per-unit tariff on its imported garments,

- A. the domestic demand for imported garments would increase.
- B. local consumers' expenditure on imported garments would increase.
- C. the sales revenue of domestic garment producers would decrease.
- D. the tariff revenue of the country coming from imported garments would decrease.

Answer Questions 11 and 12 by referring to the following market demand schedule of a good.

Price (\$)	11	12	13	14
Quantity demanded (unit)	160	140	120	100

2003/CE/II/11

Suppose the government imposes a unit sales tax of \$3 on the good. As a result, the equilibrium price of the good increases from \$12 per unit to \$13 per unit. Which of the following descriptions is correct?

- A. The consumers bear a larger tax burden than the producers.
- B. The elasticity of demand of the good is greater than 1.
- C. The total tax burden is \$120.
- D. The total sales revenue (including tax) has increased.

2003/CE/II/12

If the equilibrium price of the good falls back to \$12 per unit due to a decrease in production cost,

- A. the producers will bear all the tax burden.
- B. the total tax burden will be \$360.
- C. there will be excess demand.
- D. the total sales revenue net of tax will be \$1 260.

2003/CE/II/42

Suppose the government of Hong Kong introduces a general sales tax. It follows that

- A. the sales revenue net of tax to the sellers will decrease.
- B. there will be a persistent increase in the general price level.
- C. income will be redistributed from the rich to the poor.
- D. the consumption expenditure of Hong Kong will increase.

2005/CE/II/09

Under the Closer Economic Partnership Arrangement (CEPA), some of Hong Kong made goods exported to the Chinese mainland will enjoy zero tariffs. As a result, in the mainland,

- A. the price of these goods will rise.
- B. the price of these goods will fall.
- C. the total expenditure on these goods will rise.
- D. the total expenditure on these goods will fall.

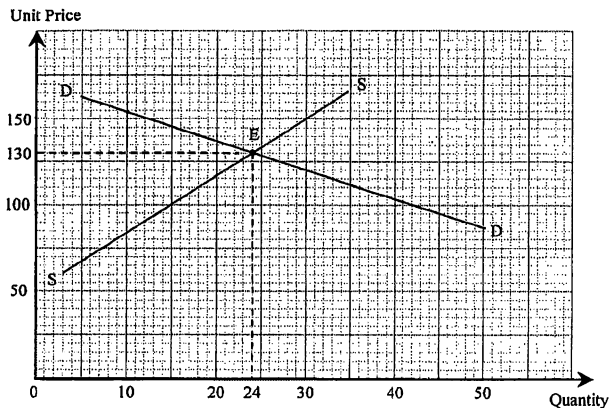
2005/CE/II/11

Good X (unit price)	Quantity demanded in the market (unit)	Quantity supplied in the market (unit)
\$6	60	40
\$9	55	45
\$12	50	50
\$15	45	55
\$18	40	60

If the government uses a unit tax instead to fix the output of Good X at 45 units, the unit tax imposed on Good X will be

- A. \$3
- B. \$6
- C. \$9
- D. \$12

Answer questions 11 and 12 by referring to the following information: The graph below shows the market situation of Good A when a per-unit sales tax of \$30 is charged on it.



2006/CE/II/11

Suppose the above per-unit sales tax is abolished. The new market price would be

- A. \$100.
- B. \$120.
- C. \$140.
- D. \$160.

2006/CE/II/12

Within the range of the original and new market prices, the elasticity of demand of Good A is

- A. smaller than its elasticity of supply.
- B. equal to its elasticity of supply.
- C. greater than its elasticity of supply.
- D. unity.

2008/CE/II/07

The following table shows the supply of and the demand for Good X.

Unit price (\$)	Quantity demanded (units)	Quantity supplied (units)
5	10	50
4	20	40
3	30	30
2	40	20
1	50	10

If the government imposes a \$2 per unit sales tax on Good X, the new equilibrium price will be _____.

- A. \$2
- B. \$3
- C. \$4
- D. \$5

2008/CE/II/08

Suppose the per unit sales tax on red wine is reduced. The sellers will enjoy a smaller proportion of a reduction in tax burden than the buyers if the demand for red wine is _____ and the supply is _____.

- A. elastic elastic
- B. elastic inelastic
- C. inelastic elastic
- D. inelastic inelastic

2009/CE/II/13

The following table shows the supply of and the demand for Good X.

Unit price (\$)	Quantity demanded (units)	Quantity supplied (units)
6	10	60
5	20	50
4	25	40
3	30	30
2	50	20
1	60	10

If the government imposes a \$3 per unit tax on Good X, the tax burden for consumers would be _____.

- A. \$20
- B. \$40
- C. \$60
- D. \$100

The following table shows the market of Good M.

Unit price	Quantity demanded (units)	Quantity supplied (units)
\$2	1 200	400
\$4	1 000	600
\$6	800	800
\$8	600	1 000
\$10	400	1 200

Suppose the government imposes a per unit sales tax of \$4 on Good M. Find the correct answers for questions 11 and 12.

2010/CE/II/11

The new market price is _____.

- A. \$4
- B. \$6
- C. \$8
- D. \$10

2010/CE/II/12

The total tax burden on producers is _____.

- A. \$1 200
- B. \$1 600
- C. \$2 400
- D. \$3 200

Microeconomics by Topic
8. Market Intervention

Refer to the following demand-supply schedule of good X and answer Questions 16 and 17.

Price (\$)	Quantity demanded (units)	Quantity supplied (units)
10	700	300
11	600	400
12	500	500
13	400	600
14	300	700
15	200	800

2012/DSE/I/16

If the government imposes a per unit tax of \$4 on good X,

- A. the total tax revenue will be \$2 000.
- B. the total sales revenue will drop by \$1 800.
- C. the demand for good X is inelastic.
- D. the total sales revenue of the substitutes of good X will drop.

2012/DSE/I/17

With reference to the above question, the ratio of the consumers' tax burden to the producers' tax burden is

- A. 1:1
- B. 1:3
- C. 3:1
- D. None of the above

2013/DSE/I/13

The following table shows the demand and supply schedules of good X.

Price (\$)	2	3	4	5	6	7	8
Quantity demanded (units)	90	80	70	60	50	40	30
Quantity supplied (units)	30	40	50	60	70	80	90

Suppose the government imposes a \$2 per-unit sales tax on each unit of good X sold. The total sales revenue net of tax is

- A. \$180
- B. \$200
- C. \$280
- D. \$300

2014/DSE/I/22

The imposition of a per-unit sales tax on cosmetics will

- A. reduce the total expenditure on cosmetics.
- B. reduce the total sales revenue (net-of-the-tax).
- C. increase the total social surplus in the cosmetics market.
- D. increase the producer surplus in the cosmetics market.

2015/DSE/I/16

Unit Price (\$)	4	5	6	7	8
Quantity demanded (units)	16	14	12	10	8
Quantity supplied (units)	14	14	14	14	14

The government imposes a \$2 per unit tax on the good. Which of the following statements about the good is correct?

- A. Sellers' revenue net of tax is \$70.
- B. The new market is \$6.
- C. The total tax revenue is \$28.
- D. The above information is insufficient to determine the tax burden on sellers.

2017/DSE/I/16

Refer to the following supply and demand schedules for Good X.

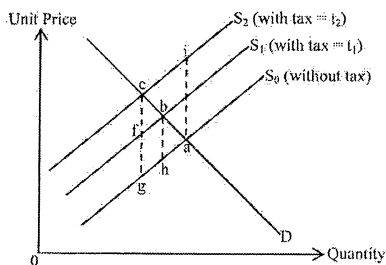
Unit Price (\$)	6	7	8	9	10	11	12	13
Quantity demanded	140	130	120	110	100	90	80	70
Quantity supplied	60	70	80	90	100	110	120	130

If the government imposed a \$2 per unit tax on Good X,

- A. sellers' revenue net of tax would become \$990.
- B. tax burden borne by the sellers would be \$90.
- C. the total tax revenue would be \$200.
- D. buyers' total expenditure on Good X would decrease by \$40.

2018/DSE/I/22

The following diagram shows the demand and supply curves of Good X.



Suppose the government increases the tax on Good X from t_1 to t_2 . Which of the following statements is correct?

- A. The deadweight loss will increase by area bcgh.
- B. The deadweight loss will decrease by area abh.
- C. The new deadweight loss is area bcf.
- D. The new deadweight loss is area adi.

1991/CE/II/26

When the tax on commodity X is doubled, its tax revenue is also doubled. This indicates that

- A. the demand for X is perfectly elastic.
- B. the demand for X is unitarily elastic.
- C. the supply of X is perfectly elastic.
- D. the supply of X is perfectly inelastic.

1992/CE/II/26

A tax on using the Cross-Harbour Tunnel will have the least effect on solving the Tunnel's congestion problem if

- A. the demand for the Tunnel service is elastic.
- B. the demand for the Tunnel service is inelastic.
- C. the supply of the Tunnel service is elastic.
- D. the demand for other transportation means of crossing the harbour is inelastic.

1995/CE/II/11

Suppose the government imposes a per unit sales tax of \$5 on good X. Under what condition will the price increase by \$5?

- A. The supply of X is unitary elastic.
- B. The supply of X is perfectly inelastic.
- C. The demand for X is perfectly inelastic.
- D. The demand for X is unitary elastic.

2002/CE/II/11

Suppose the government imposes a unit sales tax on a good. The producers' revenue after deduction of the sales tax will remain unchanged if

- A. the demand for the good is perfectly elastic.
- B. the demand for the good is perfectly inelastic.
- C. the supply of the good is perfectly elastic.
- D. the supply of the good is perfectly inelastic.

2004/CE/II/10

Suppose the government imposes a per unit sales tax on good X of which the supply is perfectly inelastic. Only _____ will bear the tax burden, and the quantity transacted will _____.

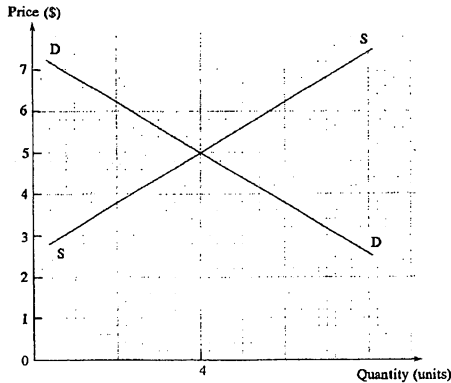
- A. sellers remain unchanged
- B. sellers decrease
- C. buyers remain unchanged
- D. buyers decrease

1990/CE/II/24

If a government abandons the policy of giving subsidy to farm products, then the price of farm products will _____ and the quantity of farm products transacted will _____.

- A. increase increase
- B. decrease decrease
- C. increase decrease
- D. decrease increase

1990/CE/II/10

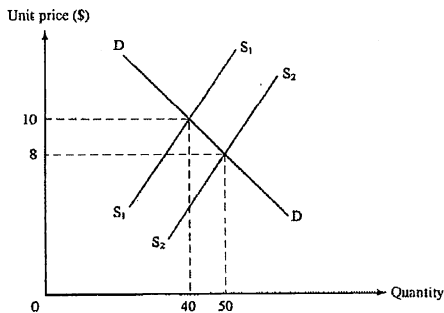


The above diagram shows the supply and demand for good X. If the government offers a subsidy of \$2 for each unit of output, the market price will

- A. decrease to \$3.
- B. decrease to \$4.
- C. increase to \$6.
- D. increase to \$7.

1996/CE/II/11

The diagram below shows that the unit price of a commodity falls from \$10 to \$8 when the government provides a per-unit subsidy of \$5 to its suppliers.



What is the total amount of suppliers' benefit from the government subsidy?

- A. \$80
- B. \$120
- C. \$150
- D. \$250

Microeconomics by Topic
8. Market Intervention

1999/CE/II/04

Case I: The market price of Good X decreases by \$4 per unit after the provision of a \$10 per unit subsidy.

Case II: The market price of Good X decreases by \$6 per unit after the provision of a \$10 per unit subsidy.

We can conclude that in comparing the two cases, Good X in Case I has a _____ demand or a _____ supply.

- A. less elastic less elastic
- B. more elastic more elastic
- C. less elastic more elastic
- D. more elastic less elastic

2000/CE/II/08

Refer to the following demand and supply schedules of a good.

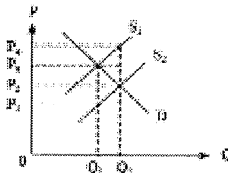
Price per unit	\$25	\$30	\$35	\$40	\$45
Quantity demanded (units)	120	100	80	75	60
Quantity supplied (units)	60	75	80	100	120

According to the above information, we can conclude that

- A. the total sales revenue would be \$3 000 when the price is \$30 per unit.
- B. a unit subsidy of \$10 on production would cause the equilibrium price to drop by \$5.
- C. a higher salaries tax would raise the market price to a price greater than \$35.
- D. the seller could set price at \$40 to gain the highest profit.

2000/CE/II/11

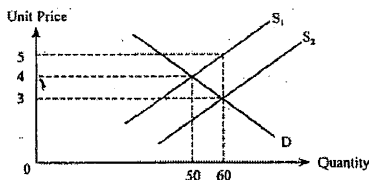
Suppose the government imposes a unit subsidy on good Y causing a shift of its supply curve from S_1 to S_2 .



The amount of subsidy benefit enjoyed by the sellers of good Y is

- A. $(P_2 - P_1) \times Q_1$
- B. $(P_3 - P_1) \times Q_1$
- C. $(P_3 - P_2) \times Q_2$
- D. $(P_4 - P_3) \times Q_2$

2001/CE/II/07



The above diagram shows the increase in supply of a good from S_1 to S_2 as a result of the provision of a per unit subsidy for the good by the government.

Suppliers' new total revenue excluding the subsidy _____ their old total revenue before the provision of the subsidy.

- A. is smaller than
- B. is equal to
- C. is larger than
- D. may be larger or smaller than

2004/CE/II/11

The following table shows the market demand and supply of a good.

Unit Price (\$)	Market quantity demanded (Units)	Market quantity supplied (Units)
10	70	130
9	90	120
8	110	110
7	130	100
6	150	90
5	170	80

If the government provides a per unit subsidy of \$3 to the suppliers,

- A. the new equilibrium price is \$5 per unit.
- B. the total amount of subsidy provided is \$390.
- C. the subsidy benefit enjoyed by the consumers is greater than that of the suppliers.
- D. the total expenditure of consumers will decrease.

2010/CE/II/09

The demand elasticity of Good X is equal to one. Which of the following is a correct description of Good X ?

- A. An increase in its price will not affect the quantity demanded.
- B. An increase in the price of its substitute will not affect the quantity demanded.
- C. The imposition of a per unit sales tax on it will not affect consumers' expenditure on it.
- D. A per unit subsidy offered by the government will not affect the total revenue (including the subsidy) of the sellers.

SP/DSE/I/20

Suppose the government provides a per-unit subsidy to the producers of a good. This would lead to

- A. an increase in consumer surplus and a fall in producer surplus, so having an ambiguous effect on efficiency.
- B. an increase in producer surplus and a fall in consumer surplus, so having an ambiguous effect on efficiency.
- C. an improvement in efficiency because both sellers and buyers would benefit from the subsidy.
- D. a deadweight loss because the amount of subsidy provided by the government is greater than the sum of the increase in consumer and producer surpluses.

2014/DSE/I/16

Refer to the following demand and supply schedules of a good.

Price	\$1	\$2	\$3	\$4	\$5
Quantity demanded (units)	120	100	80	75	60
Quantity supplied (units)	60	75	80	100	120

According to the above information, we can conclude that

- A. the total sales revenue would be \$200 when the price is \$2.
- B. there would be a shortage of 25 units when the government imposes a price floor at \$4.
- C. when a per unit subsidy of \$2 was granted to the seller, the total revenue (including the subsidy) would increase by \$160.
- D. when a per unit subsidy of \$2 was granted to the seller, the equilibrium price would drop to \$1.

2015/DSE/I/10

Which of the following will result in a rise in the total revenue of chicken farms?

- A. The price of chicken feed falls.
- B. The government reduces the profits tax rate on chicken firms.
- C. The effective quota on the sale of chickens is increased.
- D. The government reduces the subsidy for the production of pork.

2015/DSE/I/12

The government provides a per-unit subsidy of \$10 on producing green products. The market price drops by \$6 as result. Which of the following statements about the green product market is/are correct?

- (1) The supply of green products is more elastic than their demand.
 - (2) Consumers enjoy more subsidy benefits than producers.
 - (3) The demand for green products increases.
- A. (2) only
 - B. (1) and (2) only
 - C. (1) and (3) only
 - D. (1), (2) and (3)

2016/DSE/I/19

Refer to the following demand and supply schedules of a good.

Price (\$)	40	50	60	70	80	90
Quantity demanded (units)	14	12	10	8	6	4
Quantity supplied (units)	6	8	10	12	14	16

Which of the following statements is correct when the government provides a per unit subsidy of \$20 on the good?

- A. The total revenue including the subsidy is \$840.
- B. The total subsidy granted by the government is \$200.
- C. Within the range of the original and new market prices, the elasticity of demand is smaller than one.
- D. The change in total revenue of its complements is uncertain.

1994/CE/II/11

The demand for good X is perfectly inelastic. If the government offers a subsidy of \$4 for each unit of X produced, the equilibrium price of X will

- A. remain unchanged.
- B. decrease by \$4.
- C. decrease by less than \$4.
- D. decrease by more than \$4.

2020/DSE/I/15

The following table shows the demand and supply schedules of a good.

Price (\$)	Quantity demanded (units)	Quantity supplied (units)
20	70	40
22	60	45
24	50	50
26	40	55
28	30	60

If the government provides a \$6 per-unit subsidy for the producers of the good, the total subsidy provided by the government will be _____ and the ratio of consumers' share to producers' share of the subsidy will be _____.

- A. \$300.....1:1
- B. \$300.....1:2
- C. \$360.....1:1
- D. \$360.....1:2

2021/DSE/I/15

When a per-unit sales tax is imposed on a good, which of the following combinations may result in the consumers' tax burden exceeding the producers' tax burden?

- (1) elastic demand; elastic supply
 - (2) elastic demand; inelastic supply
 - (3) inelastic demand; elastic supply
 - (4) inelastic demand; inelastic supply
- A. (2) only
 - B. (1) and (3) only
 - C. (2) and (4) only
 - D. (1), (3) and (4) only

2021/DSE/I/16

Refer to the following demand-supply schedules of Good X.

Price (\$)	Quantity demanded (units)	Quantity supplied (units)
80	100	80
90	100	90
100	100	100
110	100	110
120	100	120
130	100	130

If a \$10 per-unit subsidy is provided to the producers of Good X,

- (1) the total revenue including the subsidy will remain unchanged.
- (2) the subsidy benefit will be received entirely by the producers.
- (3) the total expenditure of consumers on Good X will reduce by \$1 000.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2024/DSE/I/16

Refer to the following market demand schedule of a good. The equilibrium price is \$20.

Unit price (\$)	10	15	20	25	30	35
Quantity demanded (units)	100	90	80	70	60	50

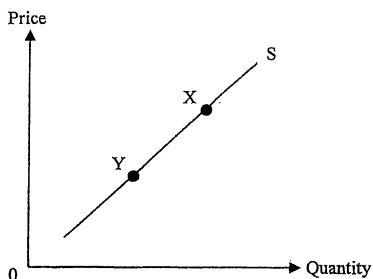
Suppose the government imposes a per-unit sales tax of \$10 on the good. At the same time, there is a decrease in the production cost of the good. As a result, the equilibrium price remains at \$20.

Based on the above information, we can conclude that _____.

- A. the consumers will not have any tax burden.
- B. producers' total revenue net of tax will be \$1 600.
- C. the deadweight loss caused by the imposition of the sales tax will disappear due to the decrease in the production cost.
- D. the government's tax revenue from the good will be \$800.

2024/DSE/I/17

The following diagram shows the supply curve of taxi services.

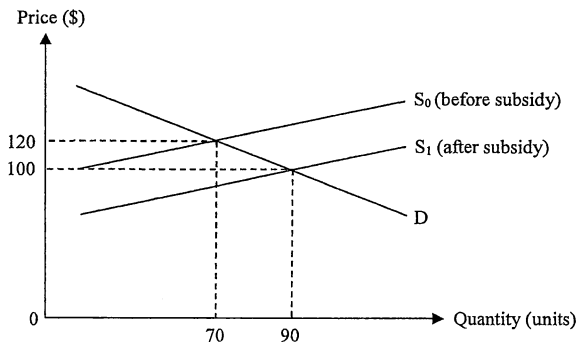


Which of the following best explains the change in the equilibrium position from point X to point Y?

- A. The government lowers the taxi fares.
- B. The government provides subsidies for citizens to use railway services.
- C. The government reduces the quota on taxi licences.
- D. The government amends the law to punish providers of ride-sharing services such as Uber drivers.

2025/DSE/I/20

The diagram below shows the market of Good X. A per-unit subsidy of \$30 is provided to the producers of Good X.



Which of the following statements about the market of Good X is correct?

- A. The subsidy benefit enjoyed by consumers is \$1 400.
- B. The total revenue inclusive of the subsidy is \$11 700.
- C. Within the range of the original and the new equilibrium prices, the price elasticity of demand is higher than the price elasticity of supply.
- D. Within the range of the original and the new equilibrium prices, the price elasticity of demand is smaller than one.

Short & Structured Questions

1991/CE/1/1(b)

In the fiscal year 90/91, the government greatly increased the tax on gasoline. With the aid of separate demand and supply diagrams, explain the effect on the equilibrium price and quantity of

- (i) gasoline, and (6 marks)
- (ii) new cars. (7 marks)

1992/CE/1/2(b)

Suppose US imposes an import tax on Hong Kong-made garments. With separate diagrams, explain the effects on the equilibrium price and quantity of

- (i) Hong Kong-made garments in the US markets, and (5 marks)
- (ii) US-made garments in the US market. (5 marks)

1993/CE/1/4(c)

Suppose a per unit tax is imposed on the TV sets sold in Hong Kong. This tax is collected from the sellers. Indicate on a supply-demand diagram the respective tax burdens of the consumers and the sellers. (5 marks)

2004/CE/1/9(c)

A first registration tax is imposed on cars and the tax is collected from car sellers. Suppose the total tax burden of car sellers is greater than that of the buyers.

- (i) State the condition that leads to this sharing of the tax burden. Illustrate the situation with a diagram.
- (ii) Using the same diagram, show car sellers' total sales revenue net of the tax. (8 marks)

2007/CE/1/9(c)

General sales tax has generated a lot of discussion.

Good X is subject to a per-unit sales tax.

- (i) With the aid of a diagram, explain under what condition consumers' total spending on Good X will decrease as a result of the introduction of the tax. (7 marks)
- (ii) Indicate in the diagram in (c)(i) the change in sellers' revenue (net of tax). (2 marks)

2008/CE/1/9(b)(ii)

The stock market of an economy is overheated. Many people withdraw money from their bank deposits to buy shares.

Two measures are suggested to cool down the overheated stock market.

- (1) the deposit interest rate of banks is raised.
- (2) the government imposes a tax on the capital gains from trading of shares.

With the aid of a diagram, explain the effect of the imposition of the capital gains tax on the total market value of shares in the economy. (7 marks)

2022/DSE/I/15

The following table shows the market supply schedule of a good.

Unit price (\$)	Quantity supplied (units)
2	22
3	26
4	30
5	34
6	38
7	42

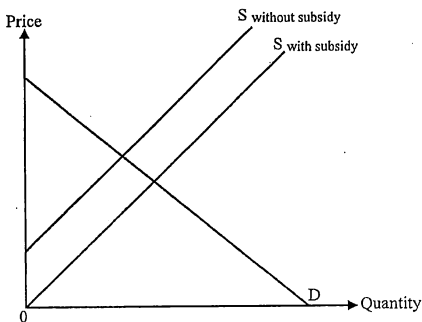
Suppose the original equilibrium price is \$5. If the government imposes a per-unit tax of \$2 on the good, the market price will rise to \$6. Which of the following statements are correct?

- (1) The ratio of buyers' tax burden to sellers' tax burden will be 1: 1.
- (2) The total tax revenue will be \$76.
- (3) The total revenue net of tax will decrease by \$50.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2022/DSE/I/21

Refer to the following supply-demand diagram of Good X.



If the government cancels the per-unit subsidy on Good X, it will lead to a decrease in ____

- (1) quantity transacted
- (2) consumer surplus
- (3) deadweight loss

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2009/CE/II/11(c)(i)

In the fiscal year 2008-09, the government took some measures to reduce the tax burden on the public.

The duties on wine (including red wine) were waived in order to establish Hong Kong as a centre for wine trading and distribution businesses.

With the fact that drinking red wine has become more popular in Hong Kong and the fact that the duty on red wine has been waived, explain in what situation the price of red wine would increase. Illustrate your answer with the aid of a supply-demand diagram. (7 marks)

2014/DSE/II/9

Many countries have considered imposing 'soft drink tax' for health reasons. Suppose a country decides to impose a per unit tax on the sellers for every bottle of soft drink sold in the market.

- (a) Under what condition will the soft drink buyers in the country bear a greater tax burden than the sellers? Illustrate your answer with a supply-demand diagram. (5 marks)
- (b) Using the same diagram in (a), indicate the change in consumer surplus after the tax is imposed. (1 mark)

2015/DSE/II/11(d)

Tobacco duty is an indirect tax in Hong Kong. The Financial Secretary has increased the tobacco duty rate with the aim of discouraging smoking.

In addition to the rise in tobacco duty, the Customs has made more effort to prevent the illegal sale of duty-not-paid cigarettes and has cracked down several illegal sellers. Is it possible that the quantity transacted of duty-not-paid cigarettes would increase? With the aid of a diagram, explain your answer. (7 marks)

2016/DSE/II/10(c)

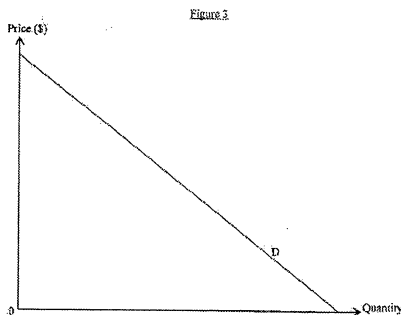
The European Union has imposed an anti-dumping tariff on the Chinese produced solar panels.

Under what condition will the producers bear a greater tariff burden than the buyers? Illustrate your answer with a supply-demand diagram. (7 marks)

2017/DSE/II/11

The government of Country X introduces a per-unit subsidy to the education services of private universities.

- (a) Under what condition would the students enjoy a larger share of subsidy benefit than the private universities? Illustrate your answer with the aid of Figure 3. (7 marks)
- (b) 'The subsidy causes inefficiency even though both students and universities benefit.' Defend the statement by comparing the marginal cost and marginal benefit in Figure 3. (3 marks)



2020/DSE/II/10B

The following source shows the current account data of the US in 2017 and 2018.

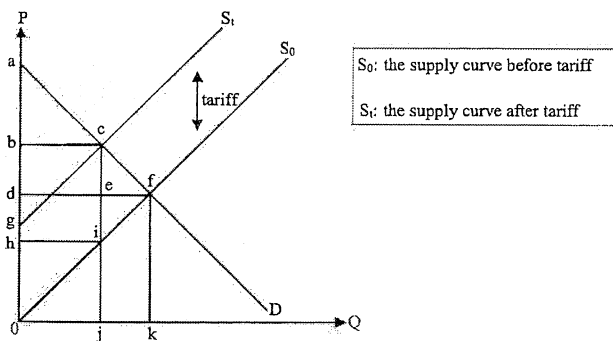
Donald Trump, the US president, once said that imposing tariffs would only increase the revenue of the US government and the US consumers did not have to pay more for the imported goods.

- (b) (i) With the aid of a supply-demand diagram, explain under **ONE** extreme condition that the consumers would bear **NO** tariff burden after the imposition of tariff. (4 marks)
- (For simplicity, the tariff is assumed to be per-unit.)
- (ii) Indicate the tariff revenue (i.e. sellers' burden) in the same diagram. (1 mark)

2020/DSE/II/10C

The following source shows the current account data of the US in 2017 and 2018.

- (c) Economists argue that tariff may harm both the consumers and producers.

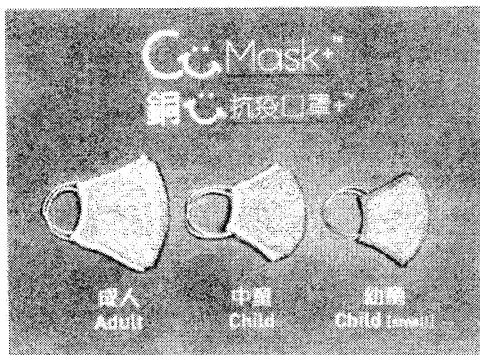


With reference to the above supply-demand diagram, complete the table below.

(3 marks)

2021/DSE/II/10c

Masks have become a daily necessity as it is compulsory for people to wear masks in many public areas.



To stabilise the supply of surgical masks in Hong Kong, the Government has launched the 'Local Mask Production Subsidy Scheme', which provides subsidy to local mask producers.

- (c) With the aid of a diagram, explain under what condition the total expenditure of consumers on surgical masks will decrease because of the subsidy.
(For simplicity, the subsidy is assumed to be a per unit one.) (7 marks)

2024/DSE/II/3

Since 2020, the government of an economy has imposed an effective quota on the production of automobiles. From 2022 to 2023, a strong economic growth was recorded in the economy.

Suppose automobiles are normal goods. How would the deadweight loss in the market of automobiles in the economy be affected by the economic growth in the above period? Explain with the aid of a diagram. (6 marks)